

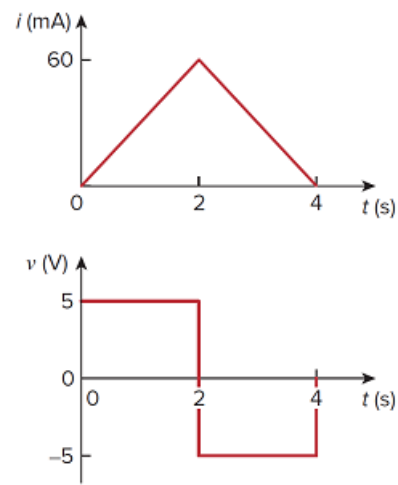
Problem

Figure 1.27 shows the current through and the voltage across an element.

(a) Sketch the power delivered to the element for $t > 0$.

(b) Find the total energy absorbed by the element for the period of $0 < t < 4$ s.

Figure 1.27:



Step-by-step solution

Step 1 of 6

Consider the current flowing through the element as shown in Figure 1:

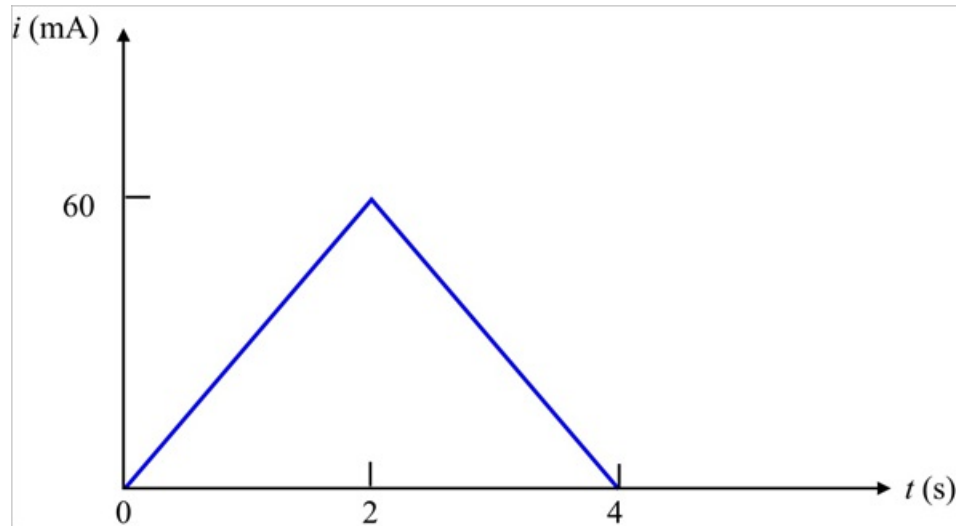


Figure 1

Step 2 of 6

Calculate the mathematical expression for Figure 1 as follows.

In the interval $0 \leq t \leq 2$:

Calculate the equation of two points $(0, 0)$ and $(2, 60)$.

$$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1} \right) (x - x_1)$$

$$i(t) - 0 = \left(\frac{60 - 0}{2 - 0} \right) (t - 0)$$

$$i(t) = 30t$$

In the interval $2 \leq t \leq 4$:

Calculate the equation of two points $(2, 60)$ and $(4, 0)$.

$$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1} \right) (x - x_1)$$

$$i(t) - 60 = \left(\frac{0 - 60}{4 - 2} \right) (t - 2)$$

$$i(t) - 60 = (-30)(t - 2)$$

$$i(t) - 60 = -30t + 60$$

$$i(t) = -30t + 120$$

The expression for current is as follows.

$$i(t) = \begin{cases} 30t \text{ mA} & \text{for } 0 \leq t \leq 2 \\ -30t + 120 \text{ mA} & \text{for } 2 \leq t \leq 4 \end{cases}$$

Step 3 of 6

Consider the voltage across the element as shown in Figure 2:

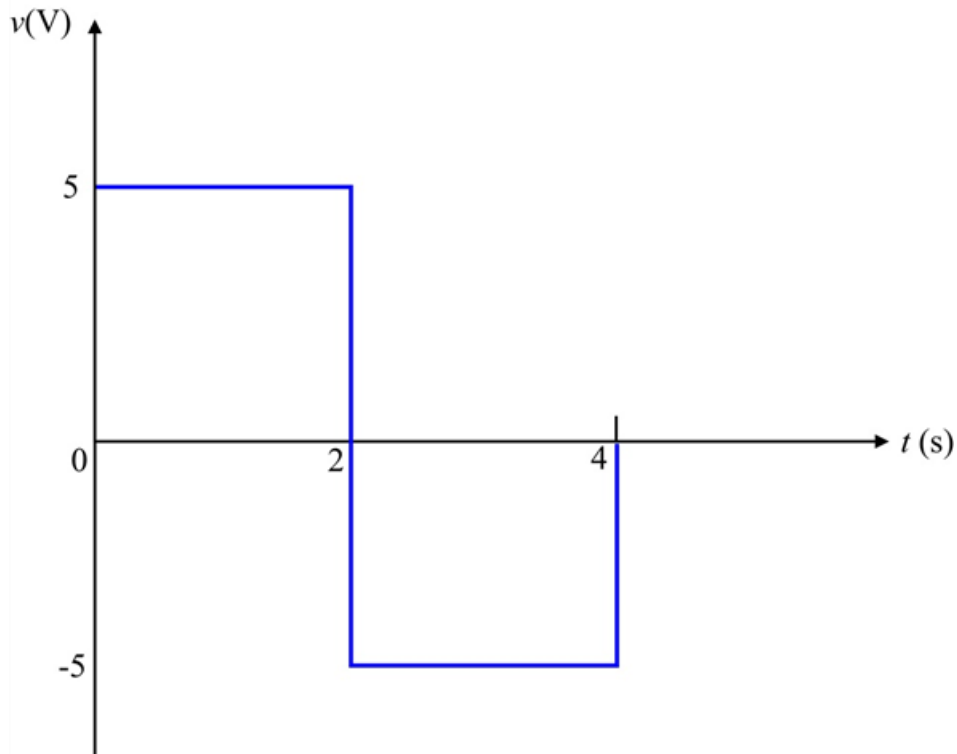


Figure 2

Step 4 of 6

Calculate the mathematical expression for Figure 2 as follows.

In the interval $0 \leq t \leq 2$:

$$v(t) = 5 \text{ V}$$

In the interval $2 \leq t \leq 4$:

$$v(t) = -5 \text{ V}$$

The expression for voltage is as follows.

$$v(t) = \begin{cases} 5 \text{ V} & \text{for } 0 \leq t \leq 2 \\ -5 \text{ V} & \text{for } 2 \leq t \leq 4 \end{cases}$$

(a)

Calculate the power delivered to the element for $t > 0$ is as follows.

Calculate the power delivered to the element in the interval $0 \leq t \leq 2$ is as follows.

$$p = vi$$

Substitute 5 for v and $30t$ for i .

$$\begin{aligned} p &= (5 \text{ V})(30t \text{ mA}) \\ &= 150t \text{ mW} \end{aligned}$$

Calculate the power delivered to the element in the interval $2 \leq t \leq 4$ is as follows.

$$p = vi$$

Substitute -5 for v and $-30t + 120$ for i .

$$\begin{aligned} p &= (-5 \text{ V})(-30t + 120 \text{ mA}) \\ &= (150t - 600) \text{ mW} \end{aligned}$$

The expression for power is as follows.

$$p(t) = \begin{cases} 150t \text{ mW} & \text{for } 0 \leq t \leq 2 \\ (150t - 600) \text{ mW} & \text{for } 2 \leq t \leq 4 \end{cases}$$

The plot of power is shown in Figure 3:

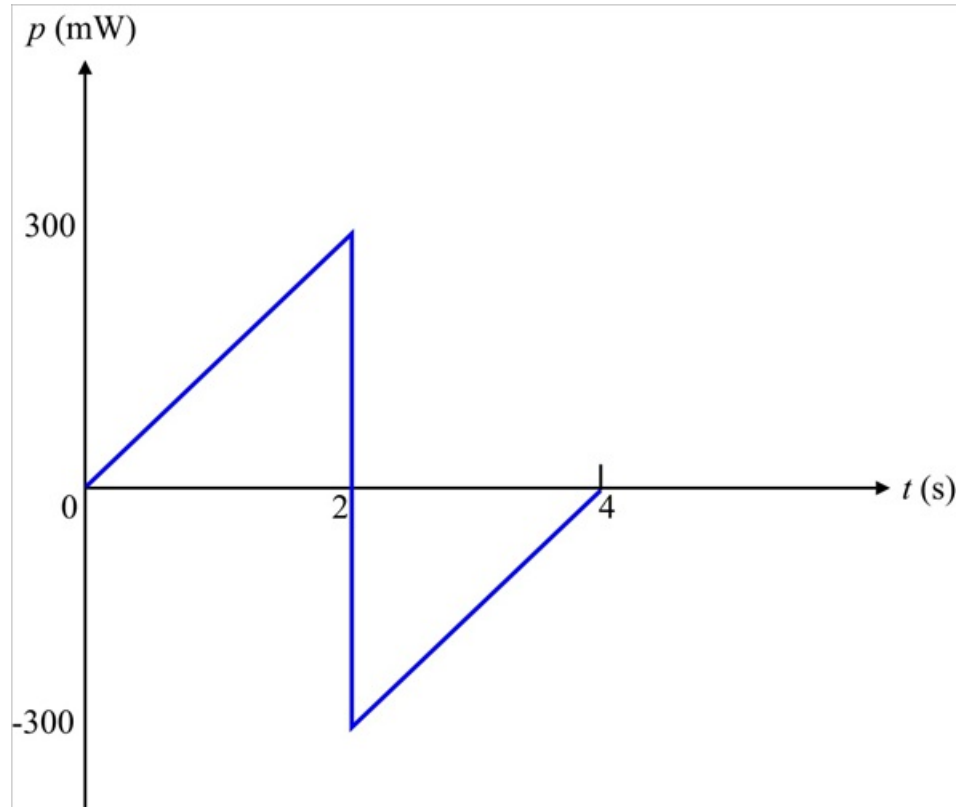


Figure 3

Step 6 of 6

(b)

Calculate the total energy absorbed by the element for the period of $0 < t < 4\text{s}$ is as follows.

$$\begin{aligned} w &= \int_{t_0}^t p \, dt \\ &= \int_0^4 p \, dt \end{aligned}$$

Substitute $150t \text{ mW}$ in the interval $0 \leq t \leq 2$ and $(150t - 600) \text{ mW}$ in the interval $2 \leq t \leq 4$

$$\begin{aligned} w &= \int_0^2 (150t) \, dt + \int_2^4 (150t - 600) \, dt \\ &= 150 \int_0^2 t \, dt + 150 \int_2^4 t \, dt - 600 \int_2^4 dt \\ &= 150 \left[\frac{t^2}{2} \right]_0^2 + 150 \left[\frac{t^2}{2} \right]_2^4 - 600 [t]_2^4 \\ &= 150 \left[\frac{4}{2} - 0 \right] + 150 \left[\frac{16}{2} - \frac{4}{2} \right] - 600 [4 - 2] \\ &= 300 + 900 - 1200 \\ &= 0 \text{ J} \end{aligned}$$

Therefore, the total energy absorbed by the element for the period of $0 < t < 4\text{s}$ is 0J.